

Quantitative Spatial Models

Building Blocks, Equilibrium, and Counterfactuals

Ningyuan Jia

University of International Business and Economics

INTRODUCTION

The Starting Question: Does X Cause Y ?

Economic theory proposes a causal relationship:

$$X \longrightarrow Y.$$

A scientific theory must confront evidence. Reduced-form analysis looks for observed variation in X while keeping other determinants of Y as unchanged as possible:

$$\Delta Y = \beta_{\text{RF}} \Delta X, \quad \beta_{\text{RF}} = \text{causal effect identified from observed variation.}$$

Reduced form answers

What happened under the observed policy change?

Policy analysis asks

What would happen under an X' that has not been observed?

Lucas critique

When the policy rule changes, expectations and decisions may also change:

$$\beta_{\text{observed regime}} \neq \beta_{\text{new regime}} \quad \text{in general.}$$

INTRODUCTION

Why Do We Need a Quantitative Model?

Reduced form estimates an effect in the observed economy. Policy analysis requires the outcome in a different economy:

$$Y = F(X, \xi, \Theta) \quad \implies \quad Y' = F(X', \xi, \Theta).$$

Here, F is the equilibrium mechanism, Θ the structural parameters, and ξ the economically interpreted baseline fundamentals.

The model makes the adjustment process explicit:

X changes $\longrightarrow$ choices change
 $\longrightarrow$ prices, wages, trade, and population adjust $\longrightarrow Y$ changes.

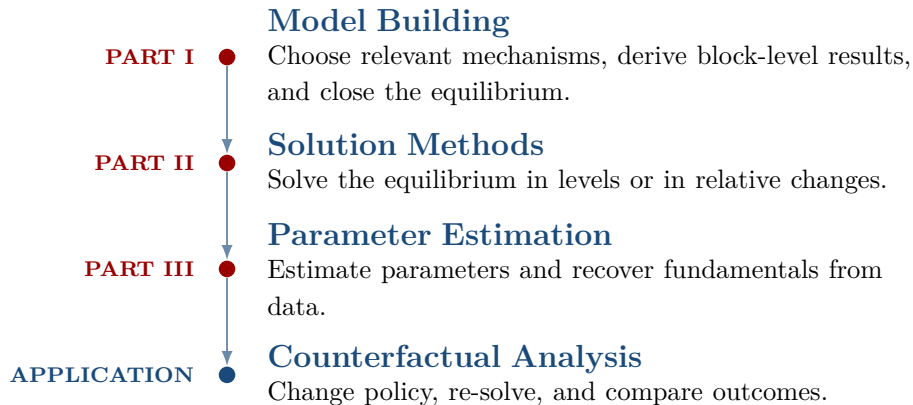
Counterfactual procedure

1. state what changes: $X \rightarrow X'$;
2. state what is held fixed: (ξ, Θ) ;
3. re-solve wages, prices, trade flows, and population, then compare Y' with Y .

The model does not reuse the old β ; it solves a new equilibrium.

INTRODUCTION

Roadmap



PART I

Model Building

PART I: MODEL BUILDING

A Menu of Model Components

Module 1: Preferences

Goods, sectors, local factors, and preference heterogeneity.

Module 2: Production Technology

Returns to scale, productivity, input–output linkages, and local factors.

Module 3: Commodity Trade

Variable or fixed trade costs, asymmetry, geography, and nontraded goods.

Module 4: Technology for Idea Flows

Knowledge diffusion, innovation, and the transferability of ideas.

Module 5: Migration

Static or dynamic location choice, migration costs, and worker heterogeneity.

Module 6: Endowments

Population and skills, housing or land, capital, and infrastructure.

Module 7: Equilibrium

Goods-market clearing, labor-market clearing, and trade balance.

The menu is open-ended

Other institutions or frictions can be added when required by the question.

A model selects one concrete specification from each relevant menu.

PART I: MODEL BUILDING

The Model Used in This Lecture

Menu	Baseline selection
Preferences	CES consumption aggregator over a single sector of differentiated goods.
Production technology	Labor-only technology with a fixed cost under monopolistic competition.
Commodity trade	CES gravity with differentiated varieties and iceberg trade costs.
Idea flows	None: location productivity is exogenous.
Migration	Static discrete choice over indirect utility, including housing congestion and idiosyncratic preference shocks.
Endowments	Homogeneous labor, fixed local housing stocks, fixed total population, and no mobile capital.
Equilibrium	General equilibrium with goods-market clearing, labor-market clearing, and trade balance.

The following slides derive selected blocks; the remaining choices stay explicit.

PART I: MODEL BUILDING

Module 1: Preferences — What does a household choose?

Consumers in location n allocate expenditure across varieties produced in origins $i \in \mathcal{N}$. Each active firm supplies one variety, and M_i is the endogenous mass of active firms in origin i :

$$C_n = \left[\sum_{i \in \mathcal{N}} \int_0^{M_i} c_{ni}(j)^{\frac{\sigma-1}{\sigma}} dj \right]^{\frac{\sigma}{\sigma-1}}, \quad \sigma > 1.$$

Let E_n denote aggregate disposable income spent on differentiated consumption goods in location n . The representative household allocates E_n across varieties:

$$\max_{\{c_{ni}(j)\}} C_n \quad \text{s.t.} \quad \sum_{i \in \mathcal{N}} \int_0^{M_i} p_{ni}(j) c_{ni}(j) dj = E_n.$$

- ▶ one sector with horizontally differentiated varieties;
- ▶ M_i is the endogenous mass of firms—and therefore varieties—in origin i ;
- ▶ σ is the elasticity of substitution across goods;

PART I: MODEL BUILDING

Module 1: Preferences — Result: Demand and the Cost of Living

Concrete equations

Demand

$$c_{ni}(j) = p_{ni}(j)^{-\sigma} P_n^{\sigma-1} E_n.$$

Price index

$$P_n = \left[\sum_{i \in \mathcal{N}} \int_0^{M_i} p_{ni}(j)^{1-\sigma} dj \right]^{1/(1-\sigma)}.$$

ϕ collects the exogenous parameters and fundamentals relevant to the module.

General form

Demand

$$\mathbf{c}_n = \mathcal{D}_n(E_n, \mathbf{p}_n, P_n; \phi),$$

Price index

$$P_n = \mathcal{P}_n(\mathbf{p}_n, \mathbf{M}; \phi).$$

→ Derivation

PART I: MODEL BUILDING

Module 2: Production Technology — How Does a Firm Produce?

A firm in location i produces one differentiated variety j :

$$\ell_i(j) = F + \frac{x_i(j)}{A_i}, \quad F > 0, \quad A_i > 0.$$

Shipping to n requires $d_{ni} \geq 1$ units at origin per unit delivered. Hence delivered marginal cost is

$$mc_{ni} = d_{ni} \frac{w_i}{A_i}.$$

- ▶ A_i : local labor productivity;
- ▶ w_i : local factor price;
- ▶ F : fixed labor requirement;
- ▶ d_{ni} : bilateral iceberg trade friction.

PART I: MODEL BUILDING

Module 2: Production Technology — Result: Delivered Prices and Firm Mass

Market structure: monopolistic competition with CES demand and free entry.

Concrete equations

Delivered price (profit maximization)

$$p_{ni} = \underbrace{\frac{\sigma}{\sigma-1}}_{\text{markup}} \underbrace{d_{ni} \frac{w_i}{A_i}}_{\text{delivered marginal cost}}.$$

Mass of firms (local labor-market clearing)

$$M_i = \frac{L_i}{\sigma F}.$$

General form

Delivered price

$$\mathbf{p}_i = \mathcal{C}_i(w_i; \phi),$$

Mass of firms

$$M_i = \mathcal{M}_i(L_i; \phi).$$

→ Derivation

PART I: MODEL BUILDING

Module 3: Commodity Trade — Result: Trade Shares

CES demand and symmetry across firms imply

Concrete equation

Trade share

General form

Trade shares

$$\begin{aligned}\pi_{ni} &\equiv \frac{\int_0^{M_i} p_{ni}(j) c_{ni}(j) dj}{\sum_k \int_0^{M_k} p_{nk}(j) c_{nk}(j) dj} \\ &= \frac{M_i p_{ni} c_{ni}}{\sum_k M_k p_{nk} c_{nk}} \\ &= \frac{M_i p_{ni}^{1-\sigma}}{\sum_k M_k p_{nk}^{1-\sigma}}.\end{aligned}$$

$$\pi_n = \mathcal{X}_n(\mathbf{p}_n, \mathbf{M}; \phi).$$

→ Derivation

PART I: MODEL BUILDING

Module 4: Technology for Idea Flows — Choices for Idea Flows

Choice	Economic content
No idea flows	Local productivity is exogenous.
Exogenous diffusion	Knowledge arriving from other locations shifts local productivity.
Endogenous innovation and diffusion	Economic activity creates ideas and affects how they spread across space.

Selection in this lecture: no idea flows

$$A_i = \bar{A}_i, \quad i \in \mathcal{N}.$$

PART I: MODEL BUILDING

Module 5: Migration — Static Discrete-Choice Migration

Location payoff. Each worker o chooses a location. Conditional on n ,

$$U_{no} = B_n \left(\frac{C_n}{\alpha} \right)^\alpha \left(\frac{h_n}{1 - \alpha} \right)^{1 - \alpha} z_{no}, \quad P_n C_n + r_n h_n = y_n + t_n.$$

Housing rents are rebated lump-sum to local residents. The local housing stock H_n is fixed and the housing market clears, so

$$t_n = \frac{r_n H_n}{L_n} = r_n h_n.$$

Idiosyncratic location preference.

$$\Pr(z_{no} \leq z) = \exp(-z^{-\kappa}), \quad \kappa > 1.$$

$y_n = E_n/L_n$ is per-capita disposable income excluding housing rents; housing rents are rebated locally; B_n is the amenity; and κ is the mobility elasticity.

PART I: MODEL BUILDING

Module 5: Migration — Migration Shares

Concrete equation

Migration share

$$\lambda_n = \frac{\left[B_n \frac{y_n}{P_n^\alpha r_n^{1-\alpha}} \right]^\kappa}{\sum_k \left[B_k \frac{y_k}{P_k^\alpha r_k^{1-\alpha}} \right]^\kappa}.$$

The Cobb–Douglas normalization cancels from location shares. This block leaves r_n as an equilibrium price; Module 7 imposes housing-market clearing.

General form

Migration share

$$\lambda = \mathcal{S}(\mathbf{y}, \mathbf{P}, \mathbf{r}; \phi).$$

→ Derivation

PART I: MODEL BUILDING

Module 6: Endowments — What Is Taken as Given?

Choices on the menu

- ▶ population and worker skills;
- ▶ housing or land endowments;
- ▶ spatial scope and geographic units;
- ▶ capital and infrastructure.

Baseline selection

- ▶ homogeneous workers;
- ▶ a fixed set of locations $\mathcal{N}$;
- ▶ fixed local housing stocks $\{H_i\}_{i \in \mathcal{N}}$;
- ▶ fixed aggregate labor $\bar{L}$;
- ▶ no mobile capital stock;
- ▶ infrastructure represented through trade costs d_{ni} .

PART I: MODEL BUILDING

Module 7: Equilibrium — Four Equilibrium Conditions

Concrete equations

Goods-market clearing

$$w_i L_i = \sum_n \pi_{ni} E_n.$$

Labor-market clearing

$$L_i = \bar{L} \lambda_i.$$

Housing-market clearing

$$H_i = L_i h_i \implies r_i = \frac{1 - \alpha}{\alpha} \frac{y_i L_i}{H_i}.$$

Trade balance

$$IM_i = (1 - \pi_{ii}) E_i,$$

$$EX_i = w_i L_i - \pi_{ii} E_i,$$

$$IM_i = EX_i + D_i \implies E_i = w_i L_i + D_i.$$

D_i is an exogenous trade deficit.

General form

Goods-market clearing

$$\mathbf{w} = \mathcal{W}(\boldsymbol{\pi}, \mathbf{E}, \mathbf{L}; \phi).$$

Labor-market clearing

$$\mathbf{L} = \mathcal{L}(\boldsymbol{\lambda}; \phi).$$

Housing-market clearing

$$\mathbf{r} = \mathcal{R}(\mathbf{y}, \mathbf{L}; \phi).$$

Trade balance

$$\mathbf{E} = \mathcal{E}(\mathbf{w}, \mathbf{L}; \phi).$$

→ Derivation

PART I: MODEL BUILDING

Module 7: Equilibrium — Assembling the Model

Unknown vector (11 blocks with 11 unknown matrices): $\mathbf{z} = (\mathbf{c}, \mathbf{p}, \mathbf{P}, \mathbf{M}, \boldsymbol{\pi}, \boldsymbol{\lambda}, \mathbf{w}, \mathbf{L}, \mathbf{y}, \mathbf{r})$.

Demand

Concrete equation

$$c_{ni} = p_{ni}^{-\sigma} P_n^{\sigma-1} E_n$$

General form

$$\mathbf{c} = \mathcal{D}(\mathbf{E}, \mathbf{p}, \mathbf{P}; \phi)$$

Price index

$$P_n = \left[\sum_i M_i p_{ni}^{1-\sigma} \right]^{1/(1-\sigma)}$$

$$\mathbf{P} = \mathcal{P}(\mathbf{p}, \mathbf{M}; \phi)$$

Delivered price

$$p_{ni} = \mu d_{ni} w_i / A_i$$

$$\mathbf{p} = \mathcal{C}(\mathbf{w}; \phi)$$

Firm mass

$$M_i = L_i / (\sigma F)$$

$$\mathbf{M} = \mathcal{M}(\mathbf{L}; \phi)$$

Trade share

$$\pi_{ni} = \frac{M_i p_{ni}^{1-\sigma}}{\sum_k M_k p_{nk}^{1-\sigma}}$$

$$\boldsymbol{\pi} = \mathcal{T}(\mathbf{p}, \mathbf{M}; \phi)$$

Migration share

$$\lambda_n = \frac{\left[B_n y_n / (P_n^\alpha r_n^{1-\alpha}) \right]^\kappa}{\sum_k \left[B_k y_k / (P_k^\alpha r_k^{1-\alpha}) \right]^\kappa}$$

$$\boldsymbol{\lambda} = \mathcal{S}(\mathbf{y}, \mathbf{P}, \mathbf{r}; \phi)$$

Disposable income

$$y_i = \frac{E_i}{L_i}$$

$$\mathbf{y} = \mathcal{Y}(\mathbf{E}, \mathbf{L}; \phi)$$

Housing rent

$$r_i = \frac{1-\alpha}{\alpha} \frac{y_i L_i}{H_i}$$

$$\mathbf{r} = \mathcal{R}(\mathbf{y}, \mathbf{L}; \phi)$$

Goods-market clearing

$$w_i L_i = \sum_n \pi_{ni} E_n$$

$$\mathbf{w} = \mathcal{W}(\boldsymbol{\pi}, \mathbf{E}, \mathbf{L}; \phi)$$

Labor-market clearing

$$L_i = \bar{L} \lambda_i$$

$$\mathbf{L} = \mathcal{L}(\boldsymbol{\lambda}; \phi)$$

Trade balance

$$E_i = w_i L_i + D_i$$

$$\mathbf{E} = \mathcal{E}(\mathbf{w}, \mathbf{L}; \phi)$$

PART I: MODEL BUILDING

Module 7: Equilibrium — Reducing the Model to Five Equation Blocks

Given fundamentals $\{A_i, B_i, H_i\}_i$, trade frictions $\{d_{ni}\}_{n,i}$, parameters $(\alpha, F, \sigma, \kappa)$, total labor $\bar{L}$, and balanced trade $\mathbf{D} = \mathbf{0}$.

Solve for $\{\mathbf{P}, \boldsymbol{\pi}, \mathbf{w}, \boldsymbol{\lambda}, \mathbf{L}\}$, with $\mu = \sigma/(\sigma - 1)$.

	Concrete equation	General form
Price index	$P_n = \mu \left[\frac{1}{\sigma F} \sum_i L_i \left(d_{ni} \frac{w_i}{A_i} \right)^{1-\sigma} \right]^{\frac{1}{1-\sigma}}$	$\mathbf{P} = \mathcal{P}(\mathbf{w}, \mathbf{L}; \phi)$
Trade share	$\pi_{ni} = \frac{L_i (d_{ni} w_i / A_i)^{1-\sigma}}{\sum_k L_k (d_{nk} w_k / A_k)^{1-\sigma}}$	$\boldsymbol{\pi} = \mathcal{T}(\mathbf{w}, \mathbf{L}; \phi)$
Goods-market clearing	$w_i L_i = \sum_n \pi_{ni} w_n L_n$	$\mathbf{w} = \mathcal{W}(\boldsymbol{\pi}, \mathbf{L}; \phi)$
Migration share	$\lambda_n = \frac{\left[B_n \frac{w_n^\alpha H_n^{1-\alpha}}{P_n^\alpha L_n^{1-\alpha}} \right]^\kappa}{\sum_k \left[B_k \frac{w_k^\alpha H_k^{1-\alpha}}{P_k^\alpha L_k^{1-\alpha}} \right]^\kappa}$	$\boldsymbol{\lambda} = \mathcal{S}(\mathbf{w}, \mathbf{P}, \mathbf{L}; \phi)$
Labor-market clearing	$L_n = \bar{L} \lambda_n$	$\mathbf{L} = \mathcal{L}(\boldsymbol{\lambda}; \phi)$
Fixed-point representation	After substituting out $\mathbf{P}$, $\boldsymbol{\pi}$, and $\boldsymbol{\lambda}$, define	

$$\mathbf{x} \equiv (\mathbf{w}, \mathbf{L}), \quad \mathbf{x} = \mathcal{F}(\mathbf{x}; \phi).$$

PART II

Solution Methods

PART II: SOLUTION METHODS

Method 1: Levels Solution — Fixed-Point Iteration

Let $k = 0, 1, 2, \dots$ denote the iteration counter.

1. **Guess.** Choose initial $\{w_i^{(0)}\}$ and $\{L_i^{(0)}\}$; use $\text{geomean}_{i \in \text{West}} w_i = 1$ as the wage numeraire.
2. **Price index (at k).** Given $(\mathbf{w}^{(k)}, \mathbf{L}^{(k)})$, compute $\mathbf{P}^{(k)}$.
3. **Shares (at k).** Compute $\boldsymbol{\pi}^{(k)}$ and $\boldsymbol{\lambda}^{(k)}$.
4. **Implied new values ($k+1$).** Update

$$L_n^{(k+1)} = \bar{L} \lambda_n^{(k)}, \quad w_i^{(k+1)} = \frac{1}{L_i^{(k)}} \sum_n w_n^{(k)} L_n^{(k)} \pi_{ni}^{(k)}.$$

5. **Compare.** If $(\mathbf{w}^{(k+1)}, \mathbf{L}^{(k+1)})$ is close to $(\mathbf{w}^{(k)}, \mathbf{L}^{(k)})$, stop.
6. **Update.** Otherwise set $k \leftarrow k+1$ and return to step 2.

PART II: SOLUTION METHODS

Method 2: Exact-Hat Algebra — The High-Dimensionality Problem

A level solution requires fundamentals, parameters, and endowments:

$$\underbrace{\{A_i, B_i, H_i\}_{i=1}^N}_{3N \text{ location fundamentals}}, \quad \underbrace{\{d_{ni}\}_{n,i=1}^N}_{N^2 \text{ bilateral trade costs}}.$$
$$\underbrace{\alpha, \sigma, \kappa, F, \{\bar{L}_c\}_c}_{\text{common parameters and country labor endowments}}.$$

With $N = 100$, bilateral trade costs alone contain 10,000 elements.

Many of these level fundamentals are unobserved. Recovering every A_i, B_i, H_i , and d_{ni} is therefore empirically demanding. In contrast, the baseline equilibrium supplies observable outcomes such as

$$\mathbf{b} = \{\mathbf{w}, \mathbf{L}, \boldsymbol{\pi}, \boldsymbol{\lambda}\}, \quad \pi_{ni} = \frac{X_{ni}}{\sum_k X_{nk}}.$$

Question: Can we conduct counterfactual policy analysis using the observed baseline equilibrium, without recovering every high-dimensional level fundamental?

PART II: SOLUTION METHODS

Method 2: Exact-Hat Algebra — What Exact-Hat Algebra Does

Counterfactual analysis usually asks how equilibrium outcomes change after a policy shock. For any object x , define

$$\widehat{x} \equiv \frac{x'}{x},$$

where x is the baseline level and x' is the counterfactual level.

Exact-hat algebra rewrites the equilibrium in proportional changes using

$$\underbrace{\{\mathbf{w}, \mathbf{L}, \boldsymbol{\pi}, \boldsymbol{\lambda}\}}_{\text{observed baseline}} + \underbrace{(\alpha, \sigma, \kappa)}_{\text{structural parameters}} + \underbrace{\{\widehat{A}, \widehat{d}, \widehat{B}, \widehat{H}, \widehat{F}, \widehat{L}\}}_{\text{shocks}}.$$

Result: solve directly for $\{\widehat{\mathbf{w}}, \widehat{\mathbf{L}}\}$. The high-dimensional levels $\{A_i, d_{ni}, B_i, H_i\}$ no longer need to be recovered separately; baseline shares become the weights in the hat system.

PART II: SOLUTION METHODS

Method 2: Exact-Hat Algebra — Exact-Hat System

Given $\mathbf{b} \equiv \{\mathbf{w}, \mathbf{L}, \boldsymbol{\pi}, \boldsymbol{\lambda}\}$ (baseline observables), $\boldsymbol{\theta} \equiv (\alpha, \sigma, \kappa)$ (structural parameters), and $\hat{\mathbf{s}} \equiv \{\hat{A}, \hat{d}, \hat{B}, \hat{H}, \hat{F}, \hat{L}\}$ (shocks). **Solve for** $\{\hat{\mathbf{P}}, \hat{\boldsymbol{\pi}}, \hat{\mathbf{w}}, \hat{\boldsymbol{\lambda}}, \hat{\mathbf{L}}\}$.

	Concrete equation	General form
Price index $\rightarrow$	$\hat{P}_n^{1-\sigma} = \hat{F}^{-1} \sum_i \pi_{ni} \hat{L}_i \left(\frac{\hat{d}_{ni} \hat{w}_i}{\hat{A}_i} \right)^{1-\sigma}$	$\hat{\mathbf{P}} = \hat{\mathcal{P}}_{\hat{\mathbf{s}}}(\hat{\mathbf{w}}, \hat{\mathbf{L}}; \mathbf{b}, \boldsymbol{\theta})$
Trade share $\rightarrow$	$\hat{\pi}_{ni} = \frac{\hat{F}^{-1} \hat{L}_i (\hat{d}_{ni} \hat{w}_i / \hat{A}_i)^{1-\sigma}}{\hat{P}_n^{1-\sigma}}$	$\hat{\boldsymbol{\pi}} = \hat{\mathcal{T}}_{\hat{\mathbf{s}}}(\hat{\mathbf{w}}, \hat{\mathbf{L}}, \hat{\mathbf{P}}; \mathbf{b}, \boldsymbol{\theta})$
Goods-market clearing $\rightarrow$	$\hat{w}_i = \frac{1}{\hat{L}_i w_i L_i} \sum_n \hat{\pi}_{ni} \hat{w}_n \hat{L}_n (w_n L_n \pi_{ni})$	$\hat{\mathbf{w}} = \hat{\mathcal{W}}_{\hat{\mathbf{s}}}(\hat{\boldsymbol{\pi}}, \hat{\mathbf{w}}, \hat{\mathbf{L}}; \mathbf{b}, \boldsymbol{\theta})$
Population share $\rightarrow$	$\hat{\lambda}_n = \frac{\left[\hat{B}_n \frac{\hat{w}_n^\alpha \hat{H}_n^{1-\alpha}}{\hat{P}_n^\alpha \hat{L}_n^{1-\alpha}} \right]^\kappa}{\sum_{k \in c} \lambda_k \left[\hat{B}_k \frac{\hat{w}_k^\alpha \hat{H}_k^{1-\alpha}}{\hat{P}_k^\alpha \hat{L}_k^{1-\alpha}} \right]^\kappa}$	$\hat{\boldsymbol{\lambda}} = \hat{\mathcal{S}}_{\hat{\mathbf{s}}}(\hat{\mathbf{w}}, \hat{\mathbf{P}}, \hat{\mathbf{L}}; \mathbf{b}, \boldsymbol{\theta})$
Labor-market clearing $\rightarrow$	$\hat{L}_n = \hat{\mathbf{L}} \hat{\lambda}_n$	$\hat{\mathbf{L}} = \hat{\mathcal{L}}_{\hat{\mathbf{s}}}(\hat{\boldsymbol{\lambda}}; \mathbf{b}, \boldsymbol{\theta})$
Fixed-point form	$\hat{\mathbf{x}} \equiv (\hat{\mathbf{w}}, \hat{\mathbf{L}}), \quad \hat{\mathbf{x}} = \hat{\mathcal{F}}_{\hat{\mathbf{s}}}(\hat{\mathbf{x}}; \mathbf{b}, \boldsymbol{\theta}),$	

$\mathbf{b} \equiv \{\mathbf{w}, \mathbf{L}, \boldsymbol{\pi}, \boldsymbol{\lambda}\}$ (baseline observables), $\boldsymbol{\theta} \equiv (\alpha, \sigma, \kappa)$ (structural parameters).

PART II: SOLUTION METHODS

Method 2: Exact-Hat Algebra — Derivation 1: Price-Index Hat

Step 1. Start from the baseline and counterfactual price indices.

$$P_n^{1-\sigma} = \frac{\mu^{1-\sigma}}{\sigma F} \sum_i L_i \left(d_{ni} \frac{w_i}{A_i} \right)^{1-\sigma}, \quad P'_n{}^{1-\sigma} = \frac{\mu^{1-\sigma}}{\sigma F'} \sum_i L'_i \left(d'_{ni} \frac{w'_i}{A'_i} \right)^{1-\sigma}.$$

Step 2. Take the ratio; the common markup term cancels.

$$\frac{P'_n{}^{1-\sigma}}{P_n^{1-\sigma}} = \widehat{F}^{-1} \frac{\sum_i L'_i (d'_{ni} w'_i / A'_i)^{1-\sigma}}{\sum_k L_k (d_{nk} w_k / A_k)^{1-\sigma}}.$$

Step 3. Substitute hats and recognize the baseline trade share.

$$\widehat{P}_n^{1-\sigma} = \widehat{F}^{-1} \sum_i \underbrace{\frac{L_i (d_{ni} w_i / A_i)^{1-\sigma}}{\sum_k L_k (d_{nk} w_k / A_k)^{1-\sigma}}}_{\pi_{ni}} \widehat{L}_i \left(\frac{\widehat{d}_{ni} \widehat{w}_i}{\widehat{A}_i} \right)^{1-\sigma}.$$

[← Back to Exact-Hat System](#)

PART II: SOLUTION METHODS

Method 2: Exact-Hat Algebra — Derivation 2: Trade-Share Hat

Step 1. Write the baseline and counterfactual trade shares.

$$\pi_{ni} = \frac{L_i(d_{ni}w_i/A_i)^{1-\sigma}}{\sum_k L_k(d_{nk}w_k/A_k)^{1-\sigma}}, \quad \pi'_{ni} = \frac{L'_i(d'_{ni}w'_i/A'_i)^{1-\sigma}}{\sum_k L'_k(d'_{nk}w'_k/A'_k)^{1-\sigma}}.$$

Step 2. Take the ratio and substitute primed objects with hats.

$$\hat{\pi}_{ni} = \frac{\hat{L}_i(\hat{d}_{ni}\hat{w}_i/\hat{A}_i)^{1-\sigma}}{\sum_k \pi_{nk} \hat{L}_k(\hat{d}_{nk}\hat{w}_k/\hat{A}_k)^{1-\sigma}}.$$

Step 3. Use the price-index hat to replace the denominator.

$$\hat{P}_n^{1-\sigma} = \hat{F}^{-1} \sum_k \pi_{nk} \hat{L}_k(\hat{d}_{nk}\hat{w}_k/\hat{A}_k)^{1-\sigma} \implies \hat{\pi}_{ni} = \frac{\hat{F}^{-1} \hat{L}_i(\hat{d}_{ni}\hat{w}_i/\hat{A}_i)^{1-\sigma}}{\hat{P}_n^{1-\sigma}}.$$

The levels A_i and d_{ni} disappear; their proportional shocks remain, while the observed baseline share π_{ni} supplies the weights.

[← Back to Exact-Hat System](#)

PART II: SOLUTION METHODS

Method 2: Exact-Hat Algebra — Derivation 3: Migration-Share Hat

Step 1. Take the hat of within-location utility.

$$V_n = \frac{w_n^\alpha H_n^{1-\alpha}}{\alpha^\alpha (1-\alpha)^{1-\alpha} P_n^\alpha L_n^{1-\alpha}} \implies \hat{V}_n = \frac{\hat{w}_n^\alpha \hat{H}_n^{1-\alpha}}{\hat{P}_n^\alpha \hat{L}_n^{1-\alpha}}.$$

Step 2. Write location shares in both equilibria.

$$\lambda_n = \frac{(B_n V_n)^\kappa}{\sum_{k \in c} (B_k V_k)^\kappa}, \quad \lambda'_n = \frac{(B'_n V'_n)^\kappa}{\sum_{k \in c} (B'_k V'_k)^\kappa}.$$

Step 3. Divide and use baseline shares as weights.

$$\hat{\lambda}_n = \frac{(\hat{B}_n \hat{V}_n)^\kappa}{\sum_{k \in c} \lambda_k (\hat{B}_k \hat{V}_k)^\kappa}.$$

Step 4. Take hats of labor-market clearing.

$$L_n = \bar{L} \lambda_n, \quad L'_n = \bar{L}' \lambda'_n \implies \hat{L}_n = \hat{\bar{L}} \hat{\lambda}_n.$$

The levels B_n, H_n disappear; only their shocks remain, while the baseline population share λ_n becomes the weight.

[← Back to Exact-Hat System](#)

PART II: SOLUTION METHODS

Method 2: Exact-Hat Algebra — Derivation 4: Wage Hat

Step 1. Write goods-market clearing in both equilibria.

$$w_i L_i = \sum_n w_n L_n \pi_{ni}, \quad w'_i L'_i = \sum_n w'_n L'_n \pi'_{ni}.$$

Step 2. Replace counterfactual levels by baseline levels times hats.

$$\widehat{w}_i \widehat{L}_i (w_i L_i) = \sum_n \widehat{\pi}_{ni} \widehat{w}_n \widehat{L}_n \underbrace{(w_n L_n \pi_{ni})}_{\text{baseline bilateral expenditure}}.$$

Step 3. Solve for the wage change.

$$\widehat{w}_i = \frac{1}{\widehat{L}_i w_i L_i} \sum_n \widehat{\pi}_{ni} \widehat{w}_n \widehat{L}_n (w_n L_n \pi_{ni}).$$

The equilibrium update uses baseline income and trade flows; it does not require separate recovery of A_i or d_{ni} .

[← Back to Exact-Hat System](#)

PART II: SOLUTION METHODS

Method 2: Exact-Hat Algebra — A 10% U.S. Tariff on China

Shock. Suppose the tariff applies to U.S. imports from China:

$$\hat{d}_{\text{US,CN}} \equiv \frac{1 + t'_{\text{US,CN}}}{1 + t_{\text{US,CN}}} = 1.10, \quad \hat{d}_{ni} = 1 \text{ otherwise,} \quad \hat{A}_i = \hat{B}_i = \hat{H}_i = \hat{F} = \hat{L} = 1.$$

The counterfactual system becomes

$$\begin{aligned}\hat{P}_n^{1-\sigma} &= \sum_i \pi_{ni} \hat{L}_i (\hat{d}_{ni} \hat{w}_i)^{1-\sigma}, \\ \hat{\pi}_{ni} &= \frac{\hat{L}_i (\hat{d}_{ni} \hat{w}_i)^{1-\sigma}}{\hat{P}_n^{1-\sigma}}, \\ \hat{w}_i &= \frac{1}{\hat{L}_i w_i L_i} \sum_n \hat{\pi}_{ni} \hat{w}_n \hat{L}_n (w_n L_n \pi_{ni}), \\ \hat{\lambda}_n &= \frac{\left[\hat{w}_n^\alpha / (\hat{P}_n^\alpha \hat{L}_n^{1-\alpha}) \right]^\kappa}{\sum_{k \in c} \lambda_k \left[\hat{w}_k^\alpha / (\hat{P}_k^\alpha \hat{L}_k^{1-\alpha}) \right]^\kappa}, \\ \hat{L}_n &= \hat{\lambda}_n.\end{aligned}$$

Exact-hat advantage: unchanged primitive hats equal one and disappear.

What must be supplied?

- ▶ baseline bilateral expenditure shares π_{ni} ;
- ▶ baseline income and population $(w_i L_i, L_i)$;
- ▶ baseline population shares λ_i ;
- ▶ structural parameters (α, σ, κ) ;
- ▶ the tariff change $\hat{d}_{\text{US,CN}}$.

What is not required?

The levels of A_i , d_{ni} , B_i , H_i , or P_i .

PART II: SOLUTION METHODS

Method 2: Exact-Hat Algebra — Counterfactual Iteration

Let $k = 0, 1, 2, \dots$ denote the iteration counter.

1. **Guess.** $\widehat{w}_i^{(0)} = \widehat{L}_i^{(0)} = 1$; impose $\text{geomean}_{i \in \text{West}} \widehat{w}_i = 1$.

2. **Price index.** Compute $\widehat{P}_n^{(k)}$ from $(\widehat{\mathbf{w}}^{(k)}, \widehat{\mathbf{L}}^{(k)})$.

3. **Shares.** Compute $\widehat{\pi}_{ni}^{(k)}$ and $\widehat{\lambda}_n^{(k)}$.

4. **Implied values.** Compute

$$\widehat{L}_n^{(k+1)} = \widehat{L} \widehat{\lambda}_n^{(k)}, \quad \widehat{w}_i^{(k+1)} = \frac{\sum_n \widehat{\pi}_{ni}^{(k)} \widehat{w}_n^{(k)} \widehat{L}_n^{(k)} (w_n L_n \pi_{ni})}{\widehat{L}_i^{(k)} w_i L_i}.$$

5. **Compare.** Stop if the new wage and population hats are sufficiently close to their previous values.

6. **Update.** Otherwise damp the implied values:

$$\widehat{w}_i^{(k+1)} \leftarrow (1 - \theta) \widehat{w}_i^{(k)} + \theta \widehat{w}_i^{(k+1)}, \quad \widehat{L}_n^{(k+1)} \leftarrow (1 - \theta) \widehat{L}_n^{(k)} + \theta \widehat{L}_n^{(k+1)}, \quad \theta \in (0, 1].$$

Set $k \leftarrow k + 1$ and return to step 2.

PART III

Parameter Estimation

PART III: PARAMETER ESTIMATION

Parameters and Fundamentals — Why Does Exact-Hat Algebra Help?

The model in levels

$$\mathbf{x} \equiv (\mathbf{w}, \mathbf{L}), \quad \mathbf{x} = \mathcal{F}(\mathbf{x}; \phi),$$

with

$$\phi = \left(\underbrace{\{A_i, B_i, H_i\}_{i=1}^N}_{\text{location fundamentals}}, \quad \underbrace{\{d_{ni}\}_{n,i=1}^N}_{\text{bilateral trade costs}}, \quad \underbrace{\alpha, F, \sigma, \kappa, \{\bar{L}_c\}_c}_{\text{common parameters and country labor endowments}} \right).$$

Recovering all elements of ϕ is demanding: many location fundamentals are unobserved, while bilateral trade costs contain N^2 elements.

The same counterfactual in changes

$$\hat{\mathbf{x}} \equiv (\hat{\mathbf{w}}, \hat{\mathbf{L}}), \quad \hat{\mathbf{x}} = \hat{\mathcal{F}}_{\hat{\mathbf{s}}}(\hat{\mathbf{x}}; \mathbf{b}, \boldsymbol{\theta}),$$

where

$$\underbrace{\mathbf{b} = \{\mathbf{w}, \mathbf{L}, \boldsymbol{\pi}, \boldsymbol{\lambda}\}}_{\text{observed baseline equilibrium}}, \quad \underbrace{\boldsymbol{\theta} = (\alpha, \sigma, \kappa)}_{\text{structural parameters}}.$$

Baseline observables replace the high-dimensional level fundamentals; only a small set of structural parameters still has to be estimated or calibrated.

PART III: PARAMETER ESTIMATION

Parameters and Fundamentals — Structural Estimation: A Common Framework

Structural estimation chooses parameters to minimize the distance between the model and the data:

$$\hat{\Theta} = \arg \min_{\Theta \in \tilde{\Theta}} \hat{\Omega}(\Theta).$$

The key choice is how to define the model–data distance $\hat{\Omega}(\Theta)$.

Model inversion

$$g^{\text{model}}(\Theta) = g^{\text{data}}.$$

Solve model equations directly for parameters or latent fundamentals.

SMM

Simulated method of moments

$$\hat{\Omega}_{\text{SMM}} = \Delta m(\Theta)' W \Delta m(\Theta),$$

$$\Delta m = m^{\text{data}} - m^{\text{sim}}(\Theta).$$

GMM

Generalized method of moments

$$\hat{\Omega}_{\text{GMM}} = \bar{g}(\Theta)' W \bar{g}(\Theta).$$

Choose parameters so empirical moment conditions are close to zero.

PART III: PARAMETER ESTIMATION

Parameters and Fundamentals — Model Inversion: Recovering the Expenditure Share

Model restriction. Household expenditure satisfies

$$P_n C_n = \alpha(y_n + t_n), \quad r_n h_n = (1 - \alpha)(y_n + t_n).$$

Hence the model can be solved directly for the expenditure parameter:

$$\alpha = \frac{P_n C_n}{P_n C_n + r_n h_n} = 1 - \frac{r_n h_n}{P_n C_n + r_n h_n}.$$

Data analogue. Aggregate expenditure accounts give

$$\hat{\alpha} = 1 - \frac{\text{housing expenditure}}{\text{goods expenditure} + \text{housing expenditure}}.$$

Example: a housing expenditure share of 25% implies $\hat{\alpha} = 0.75$. No numerical optimization is required.

PART III: PARAMETER ESTIMATION

Estimating the Trade Elasticity

Model restriction

$$\pi_{in}^j = \frac{A_i^j (c_i^j \tau_{in}^j)^{-\varepsilon_T^j}}{\sum_k A_k^j (c_k^j \tau_{kn}^j)^{-\varepsilon_T^j}}.$$

$$\tau_{in}^j = (1 + t_{in}^j) d_{in}^j.$$

Estimating equation

$$\ln X_{in}^j = -\varepsilon_T^j \ln(1 + t_{in}^j) + \alpha_i^j + \delta_n^j + \gamma_{in}^j + u_{in}^j,$$

$$\varepsilon_T^j = -\beta_{\ln(1+t)}.$$

Where it fits: IV/GMM applied to gravity moment conditions.

Examples: Caliendo and Parro (2015); Fontagné, Guimbard, and Orefice (2022).

Identification

- ▶ **Three-country ratio:** cancels exporter terms, importer terms, and symmetric bilateral costs.
- ▶ **Panel fixed effects:** absorbs supply, demand, and persistent bilateral trade-cost components.

PART III: PARAMETER ESTIMATION

Estimating the Migration Elasticity

Static discrete choice

$$m_{od} = \frac{(V_d/\mu_{od})^\kappa}{\sum_k (V_k/\mu_{ok})^\kappa},$$

$$\ln\left(\frac{m_{od}}{m_{oo}}\right) = \kappa \ln\left(\frac{V_d}{V_o}\right) - \kappa \ln\left(\frac{\mu_{od}}{\mu_{oo}}\right).$$

V_d is destination indirect utility, including income, goods prices, housing rents, and amenities.

Identification

- Fixed effects or first differences absorb persistent μ_{od} .
- Instrument V_d ; its coefficient identifies κ , then invert migration shares for μ_{od} .

Where it fits: IV estimates κ ; model inversion recovers μ_{od} .

Example: Tombe and Zhu (2019).

APPLICATION

Counterfactual Analysis

APPLICATION: COUNTERFACTUAL ANALYSIS

What Questions Does Quantitative Spatial Economics Ask?

Research question	Typical counterfactual and outcomes	Representative studies
Market access and transport infrastructure	change transport or commuting networks; measure access, urban activity, land values, congestion, and welfare	Allen & Arkolakis (2022); Donaldson & Hornbeck (2016); Donaldson (2018); Heblich et al. (2020)
Trade, technology, and sectoral shocks	change tariffs, productivity, or sectoral trade costs; measure specialization, regional incidence, and welfare	Caliendo & Parro (2015); Caliendo et al. (2019); Galle et al. (2023)
Migration, commuting, and spatial mobility	lower mobility barriers; measure population and employment reallocation, congestion, productivity, and welfare	Monte et al. (2018); Tombe & Zhu (2019)
Place-based, fiscal, and land-use policies	change subsidies, taxes, zoning, housing supply, or local density; measure sorting, externalities, spillovers, land use, rents, and welfare	Ahlfeldt et al. (2015); Fajgelbaum et al. (2019); Fajgelbaum & Gaubert (2020)
Dynamics, development, and path dependence	introduce temporary shocks or adjustment frictions; measure transitions, persistence, development, and long-run welfare	Desmet & Rossi-Hansberg (2014); Fajgelbaum & Redding (2022); Kleinman et al. (2023)

APPLICATION: COUNTERFACTUAL ANALYSIS

Exact-Hat Equilibrium Used in the Application

Given $\mathbf{b} = \{\mathbf{w}, \mathbf{L}, \boldsymbol{\pi}, \boldsymbol{\lambda}\}$, $\boldsymbol{\theta} = (\alpha, \sigma, \kappa)$, and $\hat{\mathbf{s}} = \{\hat{\mathbf{A}}, \hat{\mathbf{d}}, \hat{\mathbf{B}}, \hat{\mathbf{H}}, \hat{F}, \hat{L}\}$. **Solve for** $\{\hat{\mathbf{P}}, \hat{\boldsymbol{\pi}}, \hat{\mathbf{w}}, \hat{\boldsymbol{\lambda}}, \hat{L}\}$. Define

$$\hat{V}_n \equiv \frac{\hat{w}_n^\alpha \hat{H}_n^{1-\alpha}}{\hat{P}_n^\alpha \hat{L}_n^{1-\alpha}}.$$

Price index $\hat{P}_n^{1-\sigma} = \hat{F}^{-1} \sum_i \pi_{ni} \hat{L}_i \left(\frac{\hat{d}_{ni} \hat{w}_i}{\hat{A}_i} \right)^{1-\sigma},$

Trade share $\hat{\pi}_{ni} = \frac{\hat{F}^{-1} \hat{L}_i (\hat{d}_{ni} \hat{w}_i / \hat{A}_i)^{1-\sigma}}{\hat{P}_n^{1-\sigma}},$

Goods-market clearing $\hat{w}_i = \frac{\sum_n \hat{\pi}_{ni} \hat{w}_n \hat{L}_n (w_n L_n \pi_{ni})}{\hat{L}_i w_i L_i},$

Population share $\hat{\lambda}_n = \frac{(\hat{B}_n \hat{V}_n)^\kappa}{\sum_{k \in c} \lambda_k (\hat{B}_k \hat{V}_k)^\kappa},$

Labor-market clearing $\hat{L}_n = \hat{L}_c \hat{\lambda}_n, \quad n \in c.$

APPLICATION: COUNTERFACTUAL ANALYSIS

Inputs, Data Sources, and Environment (Redding and Rossi-Hansberg, 2017)

Input role	Exact-hat object	Empirical counterpart	Setting in the exercise
Regional observables	$\mathbf{b} : \mathbf{w}, \mathbf{L}, \boldsymbol{\lambda}$ View baseline maps	regional earnings or labor compensation; census or employment counts	synthetic regional wage and population data; 50,000 workers per country
Bilateral observable	$\mathbf{b} : \boldsymbol{\pi}$ View $\boldsymbol{\pi}$	shipment or trade-flow matrix X_{ni} , used to construct $\pi_{ni} = X_{ni} / \sum_j X_{nj}$	synthetic bilateral flows; own-location 60%, domestic nonlocal 30%, and foreign 10%
Structural parameters	$\boldsymbol{\theta} = (\alpha, \sigma, \kappa)$	expenditure accounts; trade- and migration-elasticity estimates	$\alpha = 0.75, \sigma = 5, \kappa = 1.5$
Counterfactual shocks	$\widehat{\mathbf{s}} = \{\widehat{\mathbf{A}}, \widehat{\mathbf{d}}, \widehat{\mathbf{B}}, \widehat{\mathbf{H}}, \widehat{F}, \widehat{\mathbf{L}}\}$	policy, infrastructure, productivity, amenity, housing, entry-cost, or aggregate-population changes	experiment-specific hats change; all remaining hats equal one

Simulation note. The baseline observables are simulated data generated by solving the level equilibrium for a given set of baseline parameters and fundamentals. [View baseline settings](#)

APPLICATION: COUNTERFACTUAL ANALYSIS

Counterfactual Experiments

All experiments start from the same two-country baseline economy.

Experiment	Counterfactual shock	Question
Internal trade liberalization	both countries reduce trade costs between their own regions	gains from deeper domestic integration
Bilateral border liberalization	both countries reduce international trade costs by 10%	gains from bilateral integration and regional exposure
Road construction	halve traversal costs along a West–East transport corridor	localized market-access effects and spatial spillovers
Border development zones	raise productivity by 20% in three separated West-side zones near the border	direct local gains, displacement, and cross-border spillovers

All remaining policy hats equal one in each experiment.

APPLICATION: COUNTERFACTUAL ANALYSIS

How Is Country Welfare Calculated?

The change in deterministic utility available at location n is

$$\hat{v}_n = \frac{\hat{B}_n \hat{w}_n^\alpha \hat{H}_n^{1-\alpha}}{\hat{P}_n^\alpha \hat{L}_n^{1-\alpha}}, \quad \hat{W}_c = \left[\sum_{n \in c} \lambda_n \hat{v}_n^\kappa \right]^{1/\kappa}, \quad \Delta W_c = 100 \log \hat{W}_c.$$

- ▶ λ_n is the **baseline population share within country** c ; $\sum_{n \in c} \lambda_n = 1$.
- ▶ With finite mobility elasticity $\kappa = 1.5$, $\hat{W}_c$ is the change in the discrete-choice **inclusive value**: expected utility from the full set of locations.
- ▶ It is **not** a simple population-weighted arithmetic average. For small changes, it is approximately a baseline-population-weighted average of regional log-utility changes.

APPLICATION: COUNTERFACTUAL ANALYSIS

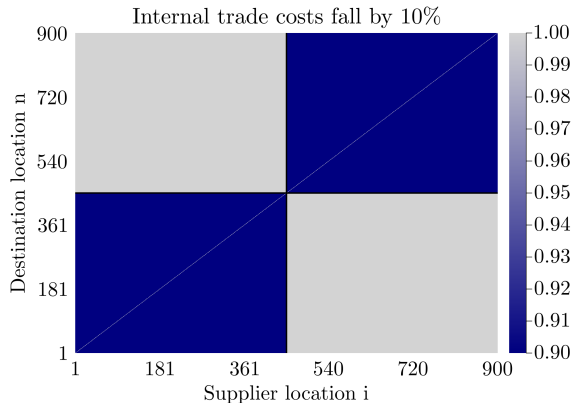
Internal Trade Liberalization: Shock

Both countries reduce internal trade costs

$$\hat{d}_{ni} = \begin{cases} 0.9, & c(n) = c(i), \ n \neq i, \\ 1, & \text{otherwise.} \end{cases}$$

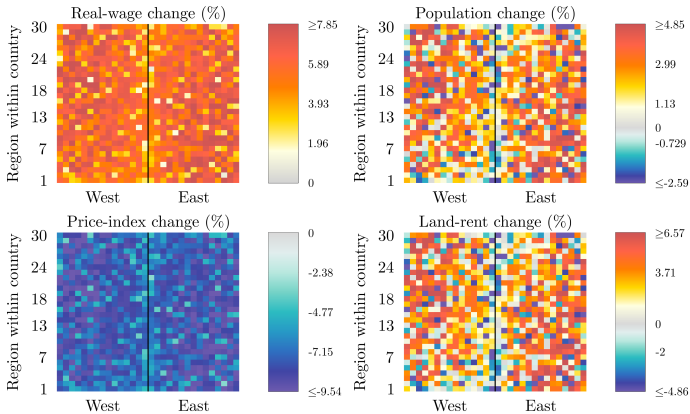
$$\hat{\mathbf{A}} = \hat{\mathbf{B}} = \hat{\mathbf{H}} = \mathbf{1}, \quad \hat{F} = \hat{\mathbf{L}} = 1.$$

Trade costs between different regions in the same country fall by 10%; cross-border pairs remain fixed.



APPLICATION: COUNTERFACTUAL ANALYSIS

Internal Trade: Results



Country inclusive-value welfare: West +4.39%; East +4.41%.
The baseline domestic interregional trade share is about 30.8%.

APPLICATION: COUNTERFACTUAL ANALYSIS

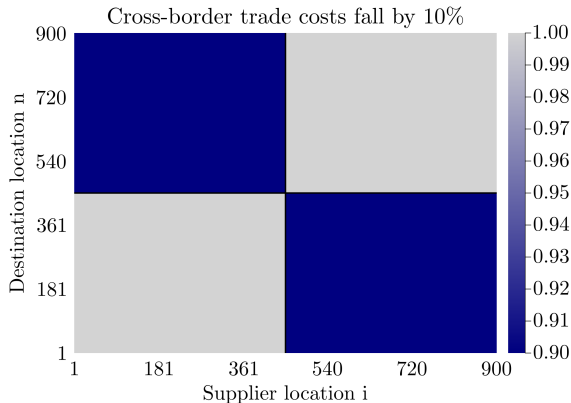
Bilateral Border Liberalization: Shock

Both countries reduce cross-border trade costs

$$\hat{d}_{ni} = \begin{cases} 0.9, & c(n) \neq c(i), \\ 1, & \text{otherwise.} \end{cases}$$

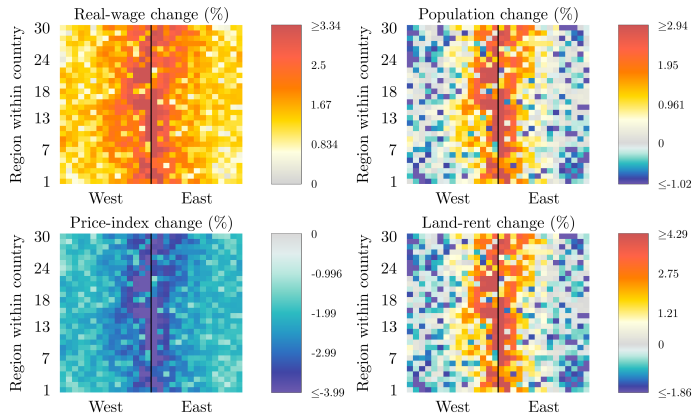
$$\hat{\mathbf{A}} = \hat{\mathbf{B}} = \hat{\mathbf{H}} = \mathbf{1}, \quad \hat{F} = \hat{\mathbf{L}} = 1.$$

Shipping costs fall by 10% in both directions across the national border. Domestic bilateral costs and all fundamentals remain fixed.



APPLICATION: COUNTERFACTUAL ANALYSIS

External Trade: Results



Country inclusive-value welfare: West +1.38%; East +1.33%.
The baseline export share is about 9.3%.

APPLICATION: COUNTERFACTUAL ANALYSIS

Road Construction: Shock

Let δ_j be the unit traversal cost of grid cell j :

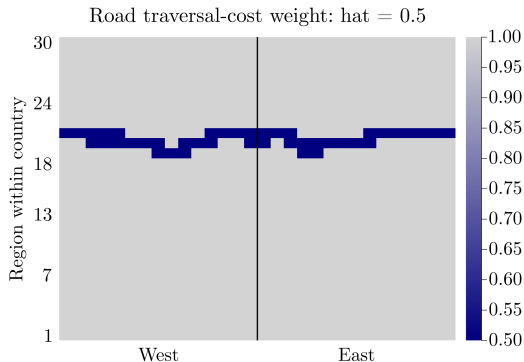
$$\hat{\delta}_j = \begin{cases} 0.50, & j \in \mathcal{R}, \\ 1, & \text{otherwise.} \end{cases}$$

Recompute the lowest-cost route and effective distance for every origin–destination pair. The matrix $\hat{\mathbf{d}}$ is then constructed from the decline in effective distance:

- ▶ a larger distance reduction implies a lower trade-cost $\hat{h}$;
- ▶ unaffected OD pairs retain $\hat{d}_{ni} = 1$.

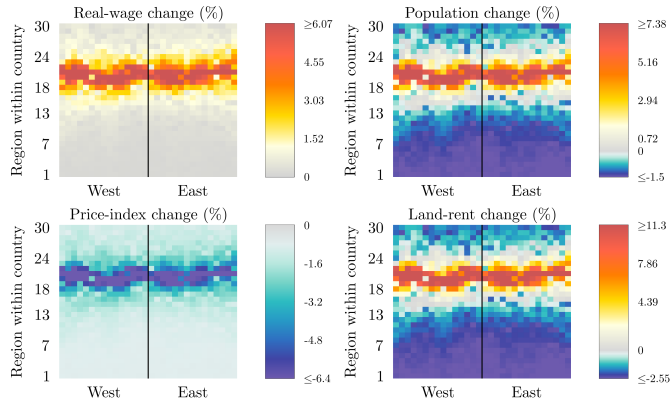
$$\hat{\mathbf{A}} = \hat{\mathbf{B}} = \hat{\mathbf{H}} = \mathbf{1}, \quad \hat{\mathbf{F}} = \hat{\mathbf{L}} = \mathbf{1}.$$

The corridor can therefore improve market access beyond the locations lying directly on the road.



APPLICATION: COUNTERFACTUAL ANALYSIS

Road Construction: Results



Country inclusive-value welfare: West +1.14%; East +1.15%.

APPLICATION: COUNTERFACTUAL ANALYSIS

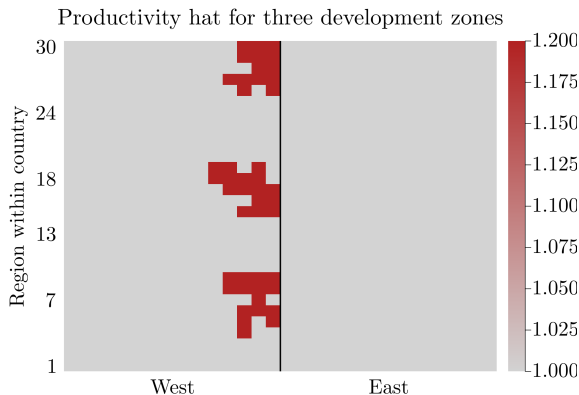
Development Zones: Shock

Let $\mathcal{Z} = \mathcal{Z}_1 \cup \mathcal{Z}_2 \cup \mathcal{Z}_3$ denote three separated development zones inside West near the border.

$$\hat{A}_i = \begin{cases} 1.20, & i \in \mathcal{Z}, \\ 1, & i \notin \mathcal{Z}. \end{cases}$$

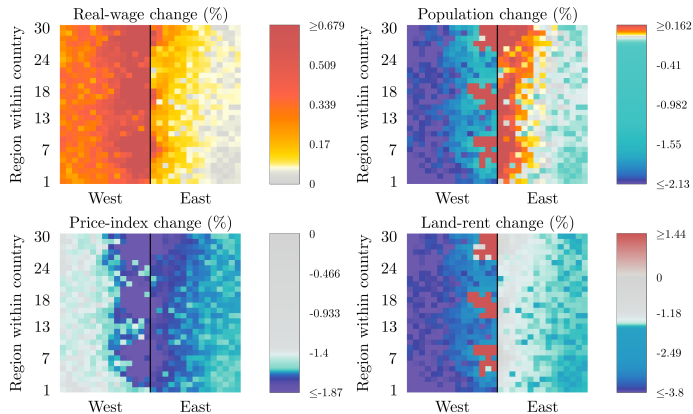
$$\hat{\mathbf{d}} = \hat{\mathbf{B}} = \hat{\mathbf{H}} = \mathbf{1}, \quad \hat{F} = \hat{\mathbf{L}} = 1.$$

The three compact zones are generated with fixed seed 20,260,725. Productivity rises by 20% only inside them; no East location is directly treated.



APPLICATION: COUNTERFACTUAL ANALYSIS

Development Zones: Results



Country inclusive-value welfare: West +1.74%; East +0.09%.
East is untreated; its smaller, spatially varying responses arise through trade spillovers.

APPLICATION: COUNTERFACTUAL ANALYSIS

Comparing the Four Policy Experiments

Counterfactual and shock	West welfare	East welfare	Mean absolute regional population change	
			West	East
internal trade liberalization $\hat{d}^{\text{dom}} = 0.9$	+4.39%	+4.41%	2.15%	2.36%
bilateral border liberalization $\hat{d}^{\text{int}} = 0.9$	+1.38%	+1.33%	0.84%	0.89%
road construction $\hat{\delta} = 0.5$ on the corridor	+1.14%	+1.15%	1.61%	1.70%
border development zones $\hat{A} = 1.2$ inside the zones	+1.74%	+0.09%	3.56%	0.08%

$$\text{Mean absolute population change in } c = \sum_{n \in c} \lambda_n 100 |\hat{L}_n - 1|.$$

Interpretation

- ▶ Both trade experiments cut affected costs by 10%. Internal liberalization applies to domestic nonlocal flows, a larger baseline expenditure share than cross-border trade.
- ▶ The corridor changes market access through the full lowest-cost-route network.
- ▶ West-side border zones directly raise local productivity and attract workers.

PART V

Extensions

PART V: EXTENSIONS

Module 1: Preferences — Extension Menu

Baseline: one-sector CES consumption, housing, exogenous amenities, and homogeneous households.

Name	Representative formulation	Economic mechanism
Multiple sectors	$C_n = \prod_j \left(\frac{C_n^j}{\alpha_n^j} \right)^{\alpha_n^j}$	Sectoral shocks, specialization, and structural transformation.
Nonhomothetic preferences	$C_n = \prod_j (C_n^j - \bar{c}_j)^{\alpha_j}$	Income-dependent expenditure shares and regional demand.
Endogenous amenities	$B_n = \bar{B}_n L_n^{\eta_B}$	City size, congestion, and place-based policies.
Quality choice	$C_n = \left[\sum_i \int (q_{ni}(j) c_{ni}(j))^{\frac{\sigma-1}{\sigma}} dj \right]^{\frac{\sigma}{\sigma-1}}$	Quality upgrading and income-dependent trade.

PART V: EXTENSIONS

Module 2: Production Technology — Extension Menu

Baseline: labor-only production with fixed costs, exogenous productivity, and no input–output linkages.

Name	Representative formulation	Economic mechanism
Constant versus increasing returns	$x_i = A_i \ell_i \quad \text{vs.} \quad \ell_i = F + \frac{x_i}{A_i}$	Firm scale, entry, and spatial concentration.
Input–output linkages	$c_i^j = \frac{1}{A_i^j} w_i^{\beta_i^j} \prod_k (P_i^k)^{\gamma_i^{kj}}$	Supply-chain propagation and local multipliers.
Multiple factors	$c_i^j = \frac{1}{A_i^j} \prod_f (w_i^f)^{\beta_{jf}}$ $A_i = \bar{A}_i \mathcal{K}_i^\eta,$	Factor intensity and distributional incidence.
Endogenous productivity	$\mathcal{K}_i = \sum_n K_{in} L_n$	Agglomeration and local-policy multipliers.
Firm heterogeneity	$\varphi \sim G_i(\varphi), \quad \Pi_{ni}(\varphi) \geq w_i f_{ni}$	Selection, export entry, and firm reallocation.
Fixed local production factors	$Y_i = A_i L_i^\beta Q_i^{1-\beta}$	Capacity constraints, rents, and specialization.

PART V: EXTENSIONS

Module 3: Commodity Trade — Extension Menu

Baseline: differentiated varieties, iceberg trade costs, one traded sector, anonymous market interaction, and no explicit transport network.

Name	Representative formulation	Economic mechanism
Eaton–Kortum sourcing	$\pi_{ni} = \frac{A_i^{EK}(d_{ni}c_i)^{-\theta}}{\sum_k A_k^{EK}(d_{nk}c_k)^{-\theta}}$	Ricardian comparative advantage and geography.
Fixed export costs	$\pi_{ni}(\omega) = \frac{r_{ni}(\omega)}{\sigma} - w_i f_{ni},$ $\text{export} \iff \pi_{ni}(\omega) \geq 0$	Export entry and the extensive trade margin.
Buyer–seller matching	$\mathcal{R}_{ni} = \chi_{ni} \mathcal{S}_i^\eta \mathcal{B}_n^{1-\eta}$	Search frictions, firm-to-firm links, and granular trade.
Transport networks	$d_{ni} = \min_{\mathcal{P}_{ni}} \prod_{(r,s) \in \mathcal{P}_{ni}} \delta_{rs}$	Roads, railways, ports, and rerouting.
Asymmetric trade costs	$d_{ni} \neq d_{in}$	Directional geography and asymmetric market access.
Nontraded goods	$d_{ni}^N = \infty, \quad n \neq i$	Local services and expenditure multipliers.

PART V: EXTENSIONS

Module 4: Technology for Idea Flows — Extension Menu

Baseline: no idea flows and exogenous local productivity.

Name	Representative formulation	Economic mechanism
Localized knowledge spillovers	$A_i = \bar{A}_i \left(\sum_n K_{in} S_n \right)^\eta$	Innovation clusters and agglomeration.
Exogenous technology diffusion	$A_{i,t+1} = (1 - \delta_A) A_{it} + \sum_n \omega_{ni} A_{nt}$	Technology diffusion and regional convergence.
Endogenous innovation	$A_{i,t+1} = (1 - \delta_A) A_{it} + \chi R_{it}^\nu$	Innovation policy and long-run growth.
Technology adoption	$A_{i,t+1} = A_{it} + \rho_i (A_t^F - A_{it})$	Catch-up and regional development.
Transferability of ideas	$A_{ni} = \chi_{ni} A_i, \quad 0 < \chi_{ni} \leq 1$	Multinational production and technology transfer.

PART V: EXTENSIONS

Module 5: Migration — Extension Menu

Baseline: static destination choice, homogeneous workers, and residence equal to workplace.

Name	Representative formulation	Economic mechanism
Bilateral migration costs	$m_{od} = \frac{(B_d V_d / \mu_{od})^\kappa}{\sum_k (B_k V_k / \mu_{ok})^\kappa}$	Migration barriers and bilateral population flows.
Dynamic migration	$V_{o,t} = u_{o,t} + \beta E_t \max_d \{ V_{d,t+1} - C_{od} + \epsilon_{od,t+1} \}$	Transition paths and adjustment speed.
Residence–workplace choice	$\lambda_{rw} = \frac{V_{rw}^\kappa}{\sum_{s,k} V_{sk}^\kappa}, \quad V_{rw} = \frac{B_r w_w}{P_r^\alpha r_r^{1-\alpha} d_{rw}}$	Commuting and local labor-market spillovers.
Worker heterogeneity	$m_{od}^g = \frac{(B_d^g V_d^g / \mu_{od}^g)^\kappa_g}{\sum_k (B_k^g V_k^g / \mu_{ok}^g)^\kappa_g}$	Sorting and heterogeneous policy incidence.
Transportation congestion	$d_{rw} = \bar{d}_{rw} \left[1 + \gamma \left(\frac{Q_{rw}}{K_{rw}} \right)^\nu \right]$	Congestion pricing and infrastructure capacity.

PART V: EXTENSIONS

Module 6: Endowments — Extension Menu

Baseline: homogeneous labor, fixed housing stocks, fixed total population, and no mobile capital.

Name	Representative formulation	Economic mechanism
Multiple skill endowments	$\sum_i L_i^g = \bar{L}^g, \quad g = 1, \dots, G$	Skill abundance, sorting, and inequality.
Elastic housing supply	$H_i = \bar{H}_i r_i^{\eta_i}$	Zoning, city growth, and rent capitalization.
Mobile capital	$R_i^K = R^K, \quad \sum_i K_i = \bar{K}$	Investment responses and regional convergence.
Capital accumulation	$\Delta K_{it} = I_{it} - \delta K_{it}$	Short-run versus long-run spatial effects.
Infrastructure endowments	$d_{ni} = d_{ni}(K_e), \quad \frac{\partial d_{ni}}{\partial K_e} < 0$	Infrastructure placement and network investment.

PART V: EXTENSIONS

Module 7: Equilibrium — Alternative Equilibrium Closures

Baseline: monopolistic competition with free entry, full general equilibrium, local rent rebates, and balanced trade.

Name	Representative formulation	Economic mechanism
Market structure	$p_i = mc_i$ vs. $p_i = \mu mc_i, \Pi_i = 0$	Markups, entry, variety, and returns to scale.
Equilibrium scope	$\mathcal{F}(\mathbf{z}^{in}; \bar{\mathbf{z}}^{out}, \phi) = 0$ vs. $\mathcal{F}(\mathbf{z}; \phi) = 0$	Endogenous versus externally fixed prices, expenditures, and utility.
Land ownership and rent distribution	$t_n L_n = \sum_i s_{ni} r_i H_i,$ $s_{ni} : \text{fixed or endogenous}$	Rent ownership, capitalization, and welfare incidence.
Trade-balance closure	$E_{it} = w_{it} L_{it} + D_{it},$ $D_{it} = \bar{D}_{it} \text{ or } (1 + R_t) a_{it} - a_{i,t+1}$	External imbalances, saving, and borrowing.

PART V: EXTENSIONS

Choosing an Extension — A Practical Checklist

Question	Modeling implication
Which adjustment margin is missing?	Select the corresponding module extension.
Which new endogenous objects appear?	Add the relevant equilibrium equations.
Which new parameters are required?	Estimate or calibrate the associated elasticities.
Which additional data are needed?	Match new flows, prices, stocks, or group-level outcomes.
Does the extension change the counterfactual result?	Retain only quantitatively important mechanisms.

PART V: EXTENSIONS

References — Core Literature and Online Resources

Core references

Redding, S. J., and E. Rossi-Hansberg. 2017. “Quantitative Spatial Economics.” *Annual Review of Economics*, 9: 21–58.

Allen, T., and C. Arkolakis. 2014. “Trade and the Topography of the Spatial Economy.” *Quarterly Journal of Economics*, 129(3): 1085–1140.

Dekle, R., J. Eaton, and S. Kortum. 2008. “Global Rebalancing with Gravity.” *IMF Staff Papers*, 55(3): 511–540.

Caliendo, L., and F. Parro. 2015. “Estimates of the Trade and Welfare Effects of NAFTA.” *Review of Economic Studies*, 82(1): 1–44.

Monte, F., S. J. Redding, and E. Rossi-Hansberg. 2018. “Commuting, Migration, and Local Employment Elasticities.” *American Economic Review*, 108(12): 3855–3890.

Caliendo, L., M. Dvorkin, and F. Parro. 2019. “Trade and Labor Market Dynamics.” *Econometrica*, 87(3): 741–835.

Lecture repository

Slides and replication code:

<https://github.com/ALLBLUEUK/structural-modeling-for-economists/tree/main/examples/quantitative-spatial-models-lecture>

Online resources

Zi Wang

Structural Models and Numerical Methods in Economics

[https:](https://github.com/WangZi-HKBU/econ7115)

[//github.com/WangZi-HKBU/econ7115](https://github.com/WangZi-HKBU/econ7115)

Treb Allen

Graduate Trade and Modern Spatial Economics

<https://sites.google.com/site/treballen/graduate-trade>

Gabriel Ahlfeldt

Lectures in Quantitative Spatial Economics

<https://github.com/Ahlfeldt/Lectures-QuantitativeSpatialEconomics>

Quantitative Urban Models

Lectures and toolkits

[https:](https://www.quantitativeurbanmodels.com/)

[//www.quantitativeurbanmodels.com/](https://www.quantitativeurbanmodels.com/)

APPENDIX

APPENDIX

Application Inputs — Baseline Settings for the Synthetic Data

Model object	Setting used in the level simulation
Geography and labor	$30 \times 30 = 900$ regions; two countries; $\bar{L}_c = 50,000$ workers in each country
Parameters	$\alpha = 0.75$, $\sigma = 5$, $\kappa = 1.5$, $F = 1$
Productivity	$\text{sd}(\log A_i) = 1.00$, normalized within country
Amenities	$B_i = 1$
Housing	fixed local stock $H_i = 100$
Iceberg trade costs	distance elasticity 0.35; nonlocal factor 2; international-border factor 2; cross-country effective-distance normalization 6.303
Equilibrium closure	fixed country populations, balanced trade, and mobility within country

$$\{\mathbf{A}, \mathbf{B}, \mathbf{H}, \mathbf{d}, F, \bar{\mathbf{L}}; \alpha, \sigma, \kappa\} \longrightarrow \{\mathbf{w}, \mathbf{L}, \mathbf{P}, \boldsymbol{\pi}, \boldsymbol{\lambda}\}.$$

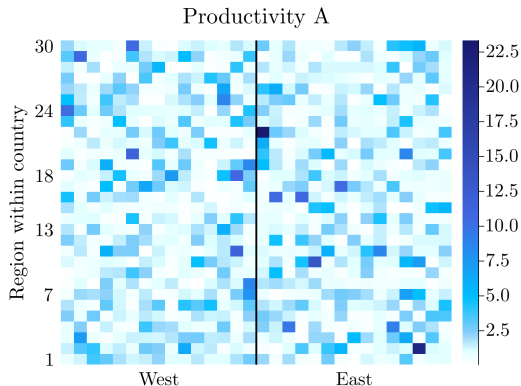
The balanced-trade simulation approximately targets own-location, domestic-nonlocal, and foreign expenditure shares of 60%, 30%, and 10% (Li et al., 2025).

[View fundamental maps](#)

[← Back to Inputs](#)

APPENDIX

Application Inputs — Baseline Regional Fundamentals

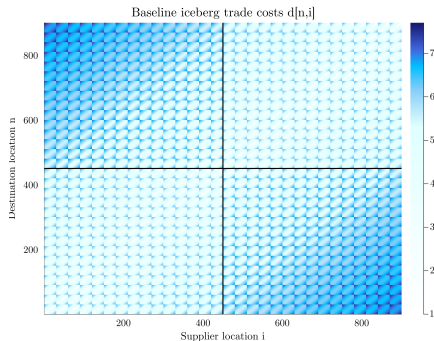


Only productivity varies across regions: $\text{sd}(\log A_i) = 1.00$, with A_i normalized within each country. The benchmark holds $B_i = 1$, $H_i = 100$, and $F = 1$ everywhere, with $\bar{L}_c = 50,000$ in each country.

[← Back to Inputs](#)

APPENDIX

Application Inputs — Iceberg Trade Costs



Direction

- i is the **origin**; n is the **destination**.
- d_{ni} is the iceberg cost of shipping a good **from region i to region n** .

Baseline setting

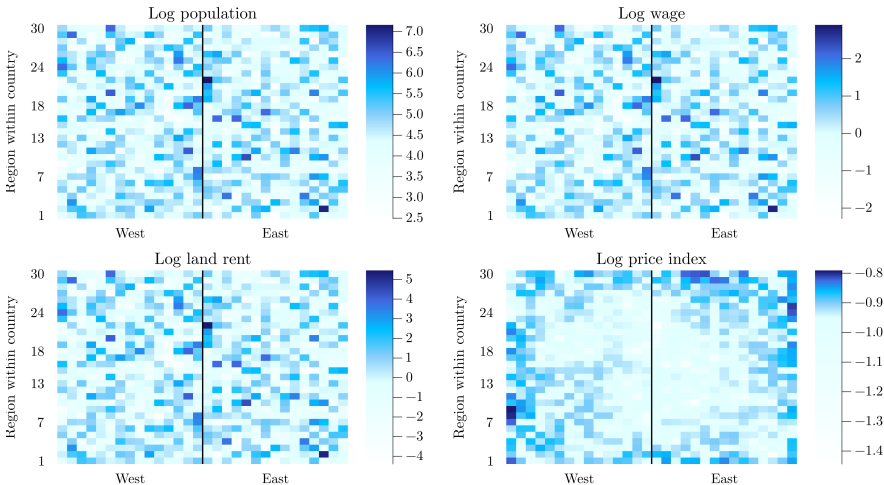
$$\begin{aligned}d_{nn} &= 1, \\d_{ni}^{\text{dom}} &= 2 \text{dist}_{ni}^{0.35}, \\d_{ni}^{\text{int}} &= 2 \times 2 \left(\frac{\text{dist}_{ni}}{6.303} \right)^{0.35}.\end{aligned}$$

The bilateral-liberalization counterfactual sets $\hat{d}_{ni} = 0.9$ whenever $c(n) \neq c(i)$. The internal-liberalization counterfactual sets $\hat{d}_{ni} = 0.9$ whenever $c(n) = c(i)$ and $n \neq i$.

← Back to Inputs

APPENDIX

Application Inputs — Baseline Equilibrium: Spatial Distribution

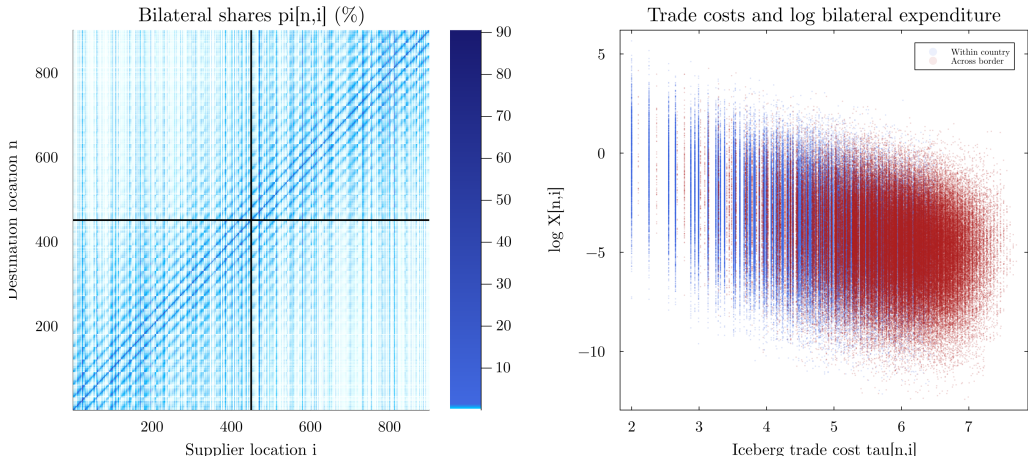


Source: Baseline observables generated by solving the level equilibrium under the simulated fundamentals.

[← Back to Inputs](#)

APPENDIX

Application Inputs — Visualizing Bilateral Expenditure Shares



Left: full π_{ni} , with destinations by row and suppliers by column. Right: $\tau_{ni} \equiv d_{ni}$ against $\log X_{ni}$, where $X_{ni} = \pi_{ni}E_n$, for $n \neq i$. The heatmap reports π_{ni} in levels with nonlinear color spacing.

[← Back to Inputs](#)

APPENDIX

Derivations — CES Demand I: Expenditure Minimization

For a target C_n , minimize expenditure:

$$\min_{\{c_{ni}(j)\}} \sum_i \int_0^{M_i} p_{ni}(j) c_{ni}(j) dj$$

subject to

$$\left[\sum_i \int_0^{M_i} c_{ni}(j)^{\frac{\sigma-1}{\sigma}} dj \right]^{\frac{\sigma}{\sigma-1}} \geq C_n.$$

The first-order conditions imply

$$\frac{c_{ni}(j)}{c_{nk}(j')} = \left[\frac{p_{ni}(j)}{p_{nk}(j')} \right]^{-\sigma}, \quad c_{ni}(j) = K_n p_{ni}(j)^{-\sigma}.$$

APPENDIX

Derivations — CES Demand II: Price Index and Marshallian Demand

Substituting $c_{ni}(j) = K_n p_{ni}(j)^{-\sigma}$ into the budget gives

$$E_n = K_n \sum_i \int_0^{M_i} p_{ni}(j)^{1-\sigma} dj.$$

The unit-expenditure function is

$$P_n = \left[\sum_i \int_0^{M_i} p_{ni}(j)^{1-\sigma} dj \right]^{1/(1-\sigma)}.$$

Therefore

$$K_n = P_n^{\sigma-1} E_n, \quad c_{ni}(j) = p_{ni}(j)^{-\sigma} P_n^{\sigma-1} E_n.$$

APPENDIX

Derivations — Markup Pricing and Free Entry

With demand $q(p) = Kp^{-\sigma}$, a firm's variable profit is

$$\pi(p) = (p - mc)Kp^{-\sigma}.$$

The first-order condition is

$$Kp^{-\sigma} - \sigma(p - mc)Kp^{-\sigma-1} = 0,$$

giving

$$p = \frac{\sigma}{\sigma - 1} mc.$$

The markup implies variable profit equals revenue divided by σ . Zero profit gives

$$\frac{R_i}{\sigma} = w_i F, \quad R_i = \sigma w_i F.$$

Variable cost is $(\sigma - 1)R_i/\sigma = (\sigma - 1)w_i F$, so aggregate labor-market clearing gives

$$L_i = M_i [F + (\sigma - 1)F] \implies M_i = \frac{L_i}{\sigma F}.$$

APPENDIX

Derivations — Trade Shares: Aggregation and Symmetry

Spending in destination n on all varieties from origin i is

$$X_{ni} = \int_0^{M_i} p_{ni}(j) c_{ni}(j) dj.$$

Therefore

$$\pi_{ni} \equiv \frac{\int_0^{M_i} p_{ni}(j) c_{ni}(j) dj}{\sum_k \int_0^{M_k} p_{nk}(j) c_{nk}(j) dj}.$$

Symmetry across firms within an origin gives

$$\pi_{ni} = \frac{M_i p_{ni} c_{ni}}{\sum_k M_k p_{nk} c_{nk}}.$$

APPENDIX

Derivations — Trade Shares: Substituting CES Demand

CES demand for a representative variety from origin i is

$$c_{ni} = p_{ni}^{-\sigma} P_n^{\sigma-1} E_n.$$

Substitution cancels the common terms $P_n^{\sigma-1} E_n$:

$$\pi_{ni} = \frac{M_i p_{ni}^{1-\sigma}}{\sum_k M_k p_{nk}^{1-\sigma}}.$$

[← Back to Module 3](#)

APPENDIX

Derivations — Housing, Income, and Indirect Utility

Let the per-capita housing-rent transfer be

$$t_n \equiv \frac{r_n H_n}{L_n}.$$

The household budget and Cobb–Douglas demands are

$$P_n C_n + r_n h_n = y_n + t_n, \quad P_n C_n = \alpha(y_n + t_n), \quad r_n h_n = (1 - \alpha)(y_n + t_n).$$

Housing-market clearing implies $H_n = L_n h_n$, so $t_n = r_n h_n$. Therefore

$$P_n C_n = y_n, \quad r_n h_n = \frac{1 - \alpha}{\alpha} y_n, \quad r_n = \frac{1 - \alpha}{\alpha} \frac{y_n L_n}{H_n}.$$

The within-location value, up to a common normalization, is

$$V_n = \frac{y_n}{P_n^\alpha r_n^{1-\alpha}}.$$

Substituting housing-market clearing gives

$$V_n = \left(\frac{\alpha}{1 - \alpha} \right)^{1-\alpha} \frac{y_n^\alpha H_n^{1-\alpha}}{P_n^\alpha L_n^{1-\alpha}}.$$

APPENDIX

Derivations — Location Choice: Shares and Expected Welfare

Let deterministic location utility be

$$v_n = B_n V_n, \quad V_n = \frac{y_n}{P_n^\alpha r_n^{1-\alpha}},$$

and let z_{no} be independent Fréchet with

$$\Pr(z_{no} \leq z) = \exp(-z^{-\kappa}).$$

Since the maximum of independent Fréchet variables is also Fréchet,

$$\Pr\left(\max_n \{v_n z_{no}\} \leq u\right) = \exp\left[-u^{-\kappa} \sum_n v_n^\kappa\right].$$

Therefore

$$\lambda_n = \frac{v_n^\kappa}{\sum_k v_k^\kappa}, \quad \mathbb{E}[U_o^*] = \Gamma\left(1 - \frac{1}{\kappa}\right) \left(\sum_n v_n^\kappa\right)^{1/\kappa}.$$

Substituting the income and housing-market equations from the preceding derivation yields the exact migration share on the main slide.

APPENDIX

Derivations — Equilibrium Closure: Market Clearing

Bilateral expenditure is $X_{ni} = \pi_{ni}E_n$. Total sales of origin i are therefore

$$R_i = \sum_n X_{ni} = \sum_n \pi_{ni}E_n.$$

With labor as the only factor and zero profits, revenue is paid to labor:

$$R_i = w_i L_i \quad \implies \quad w_i L_i = \sum_n \pi_{ni}E_n.$$

With endogenous population, labor-market clearing supplies local labor:

$$L_i = \bar{L}\lambda_i.$$

The production block has already converted local labor into the mass of firms; that condition need not be repeated here.

APPENDIX

Derivations — Equilibrium Closure: Trade Balance

Begin with the trade-balance identity

$$IM_n = EX_n + D_n.$$

Imports and exports are

$$IM_n = \sum_{i \neq n} \pi_{ni} E_n = (1 - \pi_{nn}) E_n, \quad EX_n = \sum_{m \neq n} \pi_{mn} E_m.$$

Goods-market clearing for origin n implies

$$EX_n = w_n L_n - \pi_{nn} E_n.$$

Substituting both expressions into the original identity gives

$$(1 - \pi_{nn}) E_n = w_n L_n - \pi_{nn} E_n + D_n$$

$$E_n - w_n L_n = D_n$$

$$E_n = w_n L_n + D_n.$$

Across all locations, $\sum_n D_n = 0$.

APPENDIX

Derivations — Exact-Hat Algebra: Basic Rules

For any variable x ,

$$\widehat{x} \equiv \frac{x'}{x}.$$

Products, ratios, and powers satisfy

$$\widehat{xy} = \widehat{x}\widehat{y}, \quad \widehat{x/y} = \frac{\widehat{x}}{\widehat{y}}, \quad \widehat{x^a} = (\widehat{x})^a.$$

Exact-hat algebra divides each counterfactual equilibrium equation by its baseline counterpart:

- ▶ unchanged level constants cancel;
- ▶ baseline expenditure or revenue shares become weights;
- ▶ the resulting change equations remain exact conditional on the model.

APPENDIX

Derivations — Exact-Hat Derivation: Price Index and Trade Share

Start from the two price-index equations:

$$P_n^{1-\sigma} = \frac{\mu^{1-\sigma}}{\sigma F} \sum_i L_i (d_{ni} w_i / A_i)^{1-\sigma}, \quad P_n'^{1-\sigma} = \frac{\mu'^{1-\sigma}}{\sigma F'} \sum_i L'_i (d'_{ni} w'_i / A'_i)^{1-\sigma}.$$

Dividing the second by the first and recognizing the baseline trade shares gives

$$\hat{P}_n^{1-\sigma} = \hat{F}^{-1} \sum_i \frac{L_i (d_{ni} w_i / A_i)^{1-\sigma}}{\underbrace{\sum_k L_k (d_{nk} w_k / A_k)^{1-\sigma}}_{\pi_{ni}}} \hat{L}_i (\hat{d}_{ni} \hat{w}_i / \hat{A}_i)^{1-\sigma}.$$

For trade shares, divide the counterfactual share by the baseline share:

$$\hat{\pi}_{ni} = \frac{\hat{L}_i (\hat{d}_{ni} \hat{w}_i / \hat{A}_i)^{1-\sigma}}{\sum_k \pi_{nk} \hat{L}_k (\hat{d}_{nk} \hat{w}_k / \hat{A}_k)^{1-\sigma}}.$$

The price-index equation replaces the denominator:

$$\hat{\pi}_{ni} = \frac{\hat{F}^{-1} \hat{L}_i (\hat{d}_{ni} \hat{w}_i / \hat{A}_i)^{1-\sigma}}{\hat{P}_n^{1-\sigma}}.$$

APPENDIX

Derivations — Exact-Hat Derivation: Wages and Population

Divide counterfactual goods-market clearing by baseline market clearing:

$$\begin{aligned}\widehat{w}_i \widehat{L}_i &= \sum_n \frac{w_n L_n \pi_{ni}}{w_i L_i} \widehat{\pi}_{ni} \widehat{w}_n \widehat{L}_n \\ \implies \widehat{w}_i &= \frac{1}{\widehat{L}_i w_i L_i} \sum_n \widehat{\pi}_{ni} \widehat{w}_n \widehat{L}_n (w_n L_n \pi_{ni}).\end{aligned}$$

Under balanced trade, $y_n = w_n$. After imposing housing-market clearing, the indirect-utility hat is

$$\widehat{V}_n = \frac{\widehat{w}_n^\alpha \widehat{H}_n^{1-\alpha}}{\widehat{P}_n^\alpha \widehat{L}_n^{1-\alpha}}.$$

The baseline and counterfactual migration shares are

$$\lambda_n = \frac{(B_n V_n)^\kappa}{\sum_k (B_k V_k)^\kappa}, \quad \lambda'_n = \frac{(B'_n V'_n)^\kappa}{\sum_k (B'_k V'_k)^\kappa}.$$

Dividing and collecting baseline shares gives

$$\widehat{\lambda}_n = \frac{(\widehat{B}_n \widehat{V}_n)^\kappa}{\sum_k \lambda_k (\widehat{B}_k \widehat{V}_k)^\kappa}.$$

Finally, $L_n = \bar{L} \lambda_n$ implies $\widehat{L}_n = \widehat{\bar{L}} \widehat{\lambda}_n$.